A Practical Guide to Vanishing gradient problem

TOPIC -- VANISHING GRADIENT PROBLEM

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RNN For recurrent neural networks, the long short-term memory (LSTM) network was designed to solve the problem (Hochreiter & Schmidhuber, 1997). For the exploding gradient problem, (Pascanu et al, 2012) recommended gradient clipping, meaning dividing the gradient vector g {\displaystyle g} by ■ g ■ / g max {\displaystyle \left\|g\right\|/g_{\text{max}}} if ■ g ■ > g max {\displaystyle \left\|g\right\|>g_{\text{max}}} . This restricts the gradient vectors within a ball of radius g max {\displaystyle g_{\text{max}}} .

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Δ θ = – η · [∇ x L (x T) (∇ θ F (x t – 1 , u t , θ) + ∇ x F (x t – 1 , u t , θ) ∇ θ F (x t – 2 , u t – 1 , θ) + ■)] T {\displaystyle \Delta \theta =-\eta \cdot \left[\nabla _{x}L(x_{T})\left(\nabla _{\theta }F(x_{t-1},u_{t},\theta)+\nabla _{x}F(x_{t-1},u_{t},\theta)\nabla _{\theta }F(x_{t-2},u_{t-1},\theta)+\cdots \right)\right]^{T}} where η {\displaystyle \eta } is the learning rate. The vanishing/exploding gradient problem appears because there are repeated multiplications, of the form ∇ x F (x t – 1 , u t , θ) ∇ x F (x t – 2 , u t – 1 , θ) ∇ x F (x t – 3 , u t – 2 , θ) ■ {\displaystyle \nabla _{x}F(x_{t-1},u_{t},\theta)\nabla _{x}F(x_{t-2},u_{t-1},\theta)\nabla _{x}F(x_{t-3},u_{t-2},\theta)\cdots }

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Training the network requires us to define a loss function to be minimized. Let it be L (x T , u 1 , … , u T) {\displaystyle L(x_{T},u_{1},\dots ,u_{T})} , then minimizing it by gradient descent gives

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Since | σ ′ | ≤ 1 {\displaystyle \left|\sigma '\right|\leq 1} , the operator norm of the above multiplication is bounded above by ■ W rec ■ k {\displaystyle \left\|W_{\text{rec}}\right\|^{k}} . So if the spectral radius of W rec {\displaystyle W_{\text{rec}}} is γ < 1 {\displaystyle \gamma <1} , then at large k {\displaystyle k} , the above multiplication has operator norm bounded above by γ k → 0 {\displaystyle \gamma ^{k}\to 0} . This is the prototypical vanishing gradient problem. The effect of a vanishing gradient is that the network cannot learn long-range effects. Recall Equation (loss differential): ∇ θ L = ∇ x L (x T , u 1 , … , u T) [∇ θ F (x t – 1 , u t , θ) + ∇ x F (x t – 1 , u t , θ) ∇ θ F (x t – 2 , u t – 1 , θ)

+ ■] {\displaystyle \nabla _{\theta }L=\nabla _{x}L(x_{T},u_{1},\dots ,u_{T})\left[\nabla _{\theta }F(x_{t-1},u_{t},\theta)+\nabla _{x}F(x_{t-1},u_{t},\theta)\nabla _{\theta }F(x_{t-2},u_{t-1},\theta)+\cdots \right]} The components of ∇ θ F (x , u , θ) {\displaystyle \nabla _{\theta }F(x,u,\theta)} are just components of σ (x) {\displaystyle \sigma (x)} and u {\displaystyle u} , so if u t , u t – 1 , … {\displaystyle u_{t},u_{t-1},\dots } are bounded, then ■ ∇ θ F (x t – k – 1 , u t – k , θ) ■ {\displaystyle \left\|\nabla _{\theta }F(x_{t-k-1},u_{t-k},\theta)\right\|} is also bounded by some M > 0 {\displaystyle M>0} , and so the terms in ∇ θ L {\displaystyle \nabla _{\theta }L} decay as M γ k {\displaystyle M\gamma ^{k}} . This means that, effectively, ∇ θ L {\displaystyle \nabla _{\theta }L} is affected only by the first O (γ – 1) {\displaystyle O(\gamma ^{-1})} terms in the sum. If γ ≥ 1 {\displaystyle \gamma \geq 1} , the above analysis does not quite work. For the prototypical exploding gradient problem, the next model is clearer.

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∇ x F (x t – 1 , u t , θ) ∇ x F (x t – 2 , u t – 1 , θ) ■ ∇ x F (x t – k , u t – k + 1 , θ) = W rec diag ■ (σ ′ (x t – 1)) W rec diag ■ (σ ′ (x t – 2)) ■ W rec diag ■ (σ ′ (x t – k)) {\displaystyle {\begin{aligned}&\nabla _{x}F(x_{t-1},u_{t},\theta)\nabla _{x}F(x_{t-2},u_{t-1},\theta)\cdots \nabla _{x}F(x_{t-k},u_{t-k+1},\theta)\\&=W_{\text{rec}}\operatorname {diag} (\sigma '(x_{t-1}))W_{\text{rec}}\operatorname {diag} (\sigma '(x_{t-2}))\cdots W_{\text{rec}}\operatorname {diag} (\sigma '(x_{t-k}))\end{aligned}}}

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d x d t = – x (t) + σ (w x (t) + b) + w ′ u (t) {\displaystyle {\frac {dx}{dt}}=-x(t)+\sigma (wx(t)+b)+w'u(t)}

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Other Behnke relied only on the sign of the gradient (Rprop) when training his Neural Abstraction Pyramid to solve problems like image reconstruction and face localization. Neural networks can also be optimized by using a universal search algorithm on the space of neural network's weights, e.g., random guess or more systematically genetic algorithm. This approach is not based on gradient and avoids the vanishing gradient problem.

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Consider first the autonomous case, with u = 0 {\displaystyle u=0} . Set w = 5.0 {\displaystyle w=5.0} , and vary b {\displaystyle b} in [– 3 , – 2] {\displaystyle [-3,-2]} . As b {\displaystyle b} decreases, the system has 1 stable point, then has

2 stable points and 1 unstable point, and finally has 1 stable point again. Explicitly, the stable points are (x , b) = (x , ln ■ (x 1 − x) − 5 x) {\displaystyle (x,b)=\left(x,\ln \left({\frac {x}{1-x}}\right)-5x\right)} . Now consider Δ x (T) Δ x (0) {\displaystyle {\frac {\Delta x(T)}{\Delta x(0)}}} and Δ x (T) Δ b {\displaystyle {\frac {\Delta x(T)}{\Delta b}}} , where T {\displaystyle T} is large enough that the system has settled into one of the stable points. If (x (0) , b) {\displaystyle (x(0),b)} puts the system very close to an unstable point, then a tiny variation in x (0) {\displaystyle x(0)} or b {\displaystyle b} would make x (T) {\displaystyle x(T)} move from one stable point to the other. This makes Δ x (T) Δ x (0) {\displaystyle {\frac {\Delta x(T)}{\Delta x(0)}}} and Δ x (T) Δ b {\displaystyle {\frac {\Delta x(T)}{\Delta b}}} both very large, a case of the exploding gradient. If (x (0) , b) {\displaystyle (x(0),b)} puts the system far from an unstable point, then a small variation in x (0) {\displaystyle x(0)} would have no effect on x (T) {\displaystyle x(T)} , making Δ x (T) Δ x (0) = 0 {\displaystyle {\frac {\Delta x(T)}{\Delta x(0)}}=0} , a case of the vanishing gradient. Note that in this case, Δ x (T) Δ b ≈ ∂ x (T) ∂ b = (1 x (T) (1 − x (T)) − 5) − 1 {\displaystyle {\frac {\Delta x(T)}{\Delta b}}\approx {\frac {\partial x(T)}{\partial b}}=\left({\frac {1}{x(T)(1-x(T))}}-5\right)^{-1}} neither decays to zero nor blows up to infinity. Indeed, it's the only well-behaved gradient, which explains why early researches focused on learning or designing recurrent networks systems that could perform long-ranged computations (such as outputting the first input it sees at the very end of an episode) by shaping its stable attractors. For the general case, the intuition still holds (Figures 3, 4, and 5).